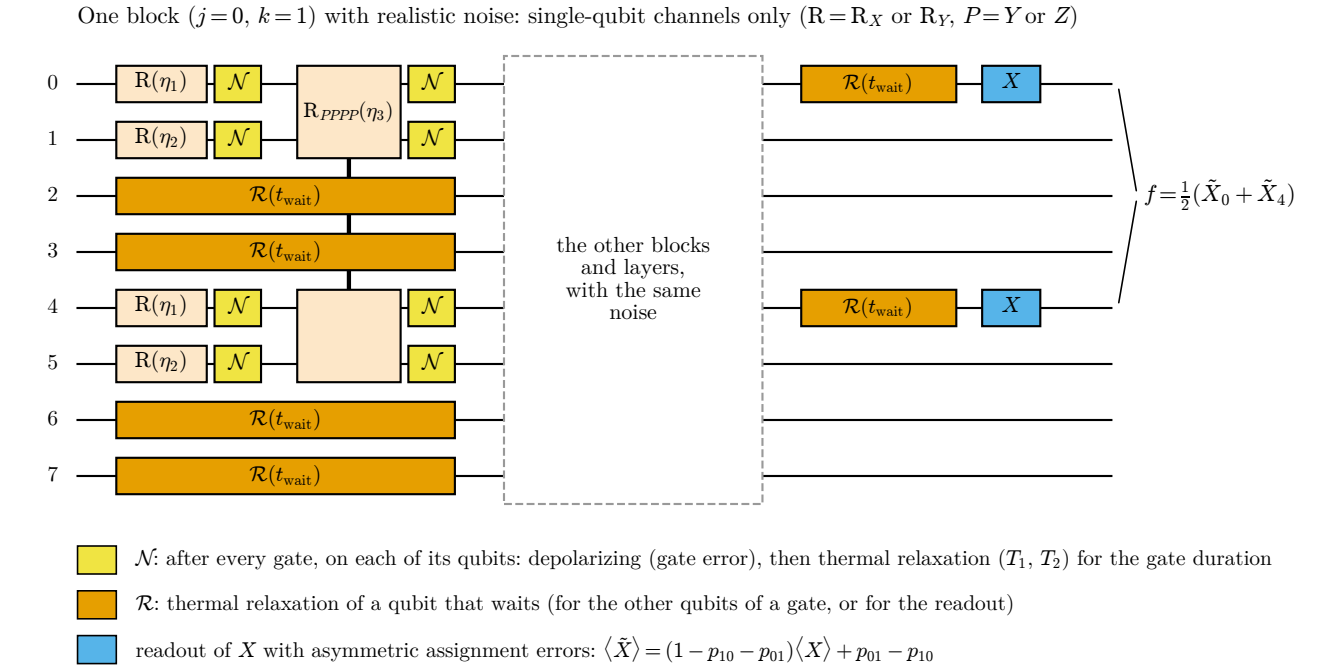
One block ($j=0$, $k=1$) with realistic noise: single-qubit channels only ($\mathrm{R}=\mathrm{R}_X$ or R_Y, $P=Y$ or Z)
0
1
2
3
4
5
6
7
$\mathrm{R}(\eta_1)$
$\mathrm{R}(\eta_2)$
$\mathrm{R}_{PPPP}(\eta_3)$
$\mathcal{N}$
$\mathcal{R}(t_{\mathrm{wait}})$
the other blocks and layers, with the same noise
X
$f=\frac{1}{2}(\tilde{X}_0+\tilde{X}_4)$
$\mathcal{N}$: after every gate, on each of its qubits: depolarizing (gate error), then thermal relaxation (T_1, T_2) for the gate duration
$\mathcal{R}$: thermal relaxation of a qubit that waits (for the other qubits of a gate, or for the readout)
readout of X with asymmetric assignment errors: $\langle\tilde{X}\rangle=(1-p_{10}-p_{01})\langle X\rangle+p_{01}-p_{10}$